

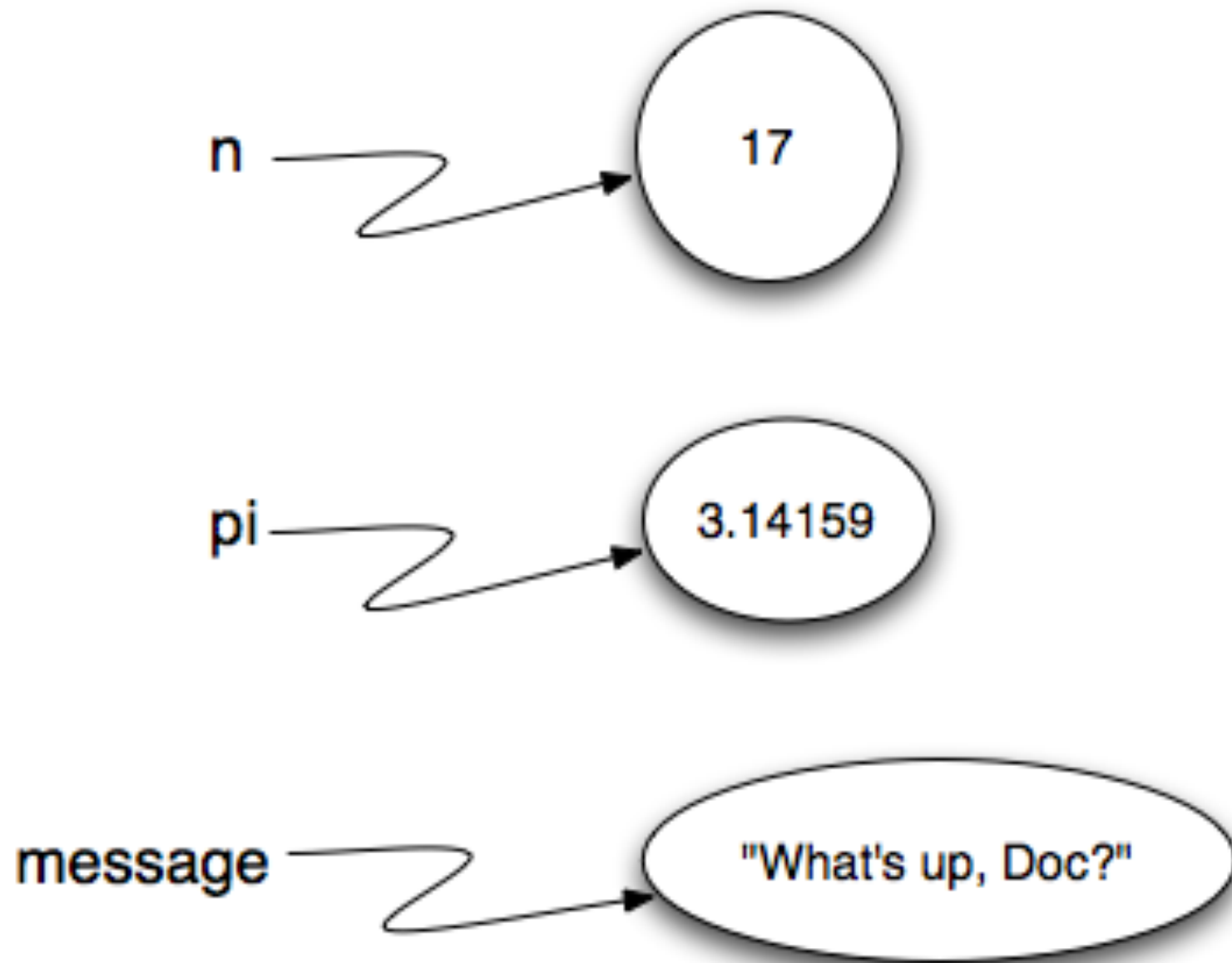
Variable er navne

```
n = 17
```

```
pi = 3.1415
```

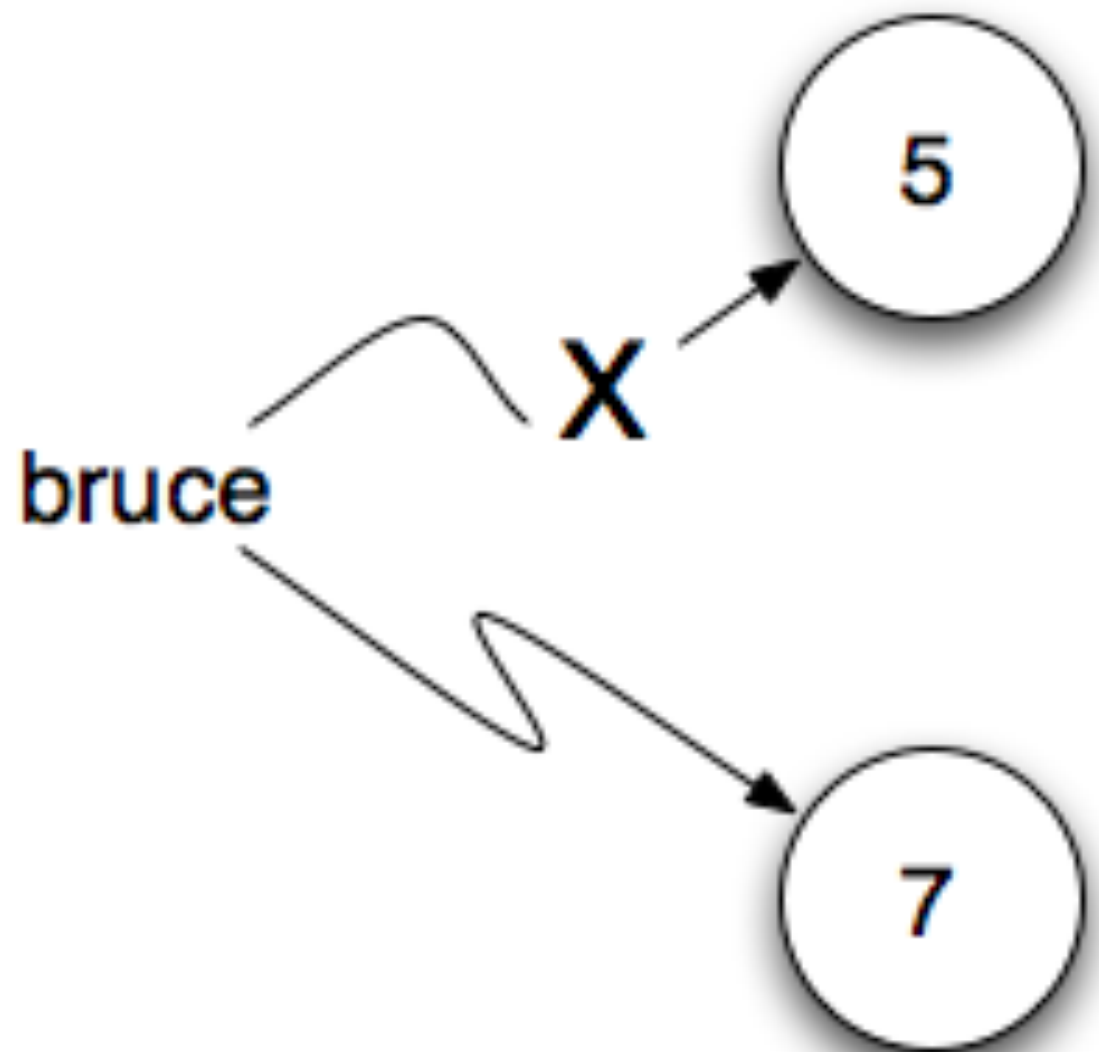
```
message = "What's up doc?"
```

Variable er navne



Ny værdi til en variabel

```
bruce = 5  
bruce = 7
```



Expressions

```
x = 2
```

```
y = 9
```

```
print(3 + x * 4 + y)
```

Expressions

`x = 2`

`y = 9`

`print(3 + x * 4 + y)`



`3 + 2 * 4 + y`

Substitution

Expressions

`x = 2`

`y = 9`

`print(3 + x * 4 + y)`

↓ ↓ ↓ ↓

3 + 2 * 4 + y

↓ ↓ ↓ ↓

3 + 8 + y

Substitution

Reduktion

Expressions

`x = 2`

`y = 9`

`print(3 + x * 4 + y)`

↓ ↓ ↓ ↓

Substitution

3 + 2 * 4 + y

↓ ↓ ↓

Reduktion

3 + 8 + y

↓ ↓ ↓

Substitution

3 + 8 + 9

Expressions

`x = 2`

`y = 9`

`print(3 + x * 4 + y)`

↓ ↓ ↓ ↓

Substitution

3 + 2 * 4 + y

↓ ↓ ↓

Reduktion

3 + 8 + y

↓ ↓ ↓

Substitution

3 + 8 + 9

↓

20



Tanketjek



Tanketjek

$$a = 1$$



Tanketjek

$$a = 1$$

$$b = a + 2$$



Tanketjek

```
a = 1
```

```
b = a + 2
```

```
print(b)
```



Tanketjek

```
a = 1
```

```
b = a + 2
```

```
print(b)
```

```
a = 8
```



Tanketjek

```
a = 1
```

```
b = a + 2
```

```
print(b)
```

```
a = 8
```

```
print(b)
```

Læg én til en variabel

```
x = 1
```

```
print(x)
```

```
x = *expression*
```

```
print(x)
```




Læg én til en variabel

```
x = 1
```

```
print(x)
```

```
x = x + 1
```

```
print(x)
```


Statements og Expressions

- **Statement:** en *line* kode der *gør noget*.
- **Expression:** en *bid* kode i en line der *reducerer til en enkelt værdi*.

Statements

```
x = 117
```

```
print(x)
```

Statements

➔ `x = 117`

`print(x)`

Statements

```
x = 117
```

```
➔ print(x)
```

Statements

```
x = 117
```

➔ `print(x)`

↓

117

Expressions

3 + 2 * 4 + 9

Expressions

```
x = 2
```

```
y = 9
```

```
print(3 + x * 4 + y)
```


Expressions

`x = 2`

`y = 9`

`print(3 + x * 4 + y)`



`3 + 2 * 4 + y`

Substitution

Expressions

`x = 2`

`y = 9`

`print(3 + x * 4 + y)`

↓ ↓ ↓ ↓

3 + 2 * 4 + y

↓ ↓ ↓ ↓

3 + 8 + y

Substitution

Reduktion

Expressions

`x = 2`

`y = 9`

`print(3 + x * 4 + y)`

↓ ↓ ↓ ↓

Substitution

3 + 2 * 4 + y

↓ ↓ ↓

Reduktion

3 + 8 + y

↓ ↓ ↓

Substitution

3 + 8 + 9

Expressions

`x = 2`

`y = 9`

`print(3 + x * 4 + y)`

↓ ↓ ↓ ↓

Substitution

3 + 2 * 4 + y

↓ ↓ ↓

Reduktion

3 + 8 + y

↓ ↓ ↓

Substitution

3 + 8 + 9

↓

20



Tanketjek



Tanketjek

$$a = 1$$



Tanketjek

$$a = 1$$

$$b = a + 2$$



Tanketjek

```
a = 1
```

```
b = a + 2
```

```
print(b)
```




Tanketjek

```
a = 1
```

```
b = a + 2
```

```
print(b)
```

```
a = 8
```




Tanketjek

```
a = 1
```

```
b = a + 2
```

```
print(b)
```

```
a = 8
```

```
print(b)
```



Læg én til en variabel

```
x = 1
```

```
print(x)
```

```
x = *expression*
```

```
print(x)
```



Læg én til en variabel

```
x = 1
```

```
print(x)
```

```
x = x + 1
```

```
print(x)
```

Relationelle operatører

= ens

!= ikke ens

> større end

< mindre end

>= større end

<= mindre end

Operator precedence

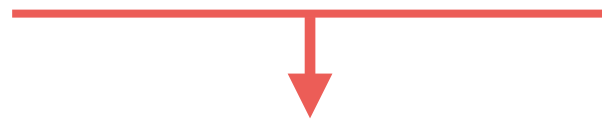
Level	Category	Operators
Highest	exponent	**
	multiplication	*, /, //, %
	addition	+, -
Lowest	relational	!=, ==, <=, >=, <, >

Relationelle operatorer

42 >= 7

Relationelle operatorer

42 >= 7



True

Relationelle operatorer

$42 \geq 7$

`True`

Relationelle operatorer producerer en værdi der enten er `True` eller `False`

Booleans

True

False

Expressions med relationelle operatorer

```
print(42 >= 7)
```

Expressions med relationelle operatorer

```
print ( 42 >= 7 )
```



True

Expressions med relationelle operatorer

```
nr = 42
```

```
print(nr >= 7)
```

Expressions med relationelle operatorer

```
nr = 42
```

```
print(nr >= 7)
```



```
42 >= 7
```

substitution

Expressions med relationelle operatorer

```
nr = 42
```

```
print(nr >= 7)
```

 
42 >= 7

substitution



reduktion

True

Expressions med relationelle operatorer

```
cars = 2
```

```
people = 7
```

```
print(cars * 4 >= people)
```


Expressions med relationelle operatorer

```
cars = 2
```

```
people = 7
```

```
print(cars * 4 >= people)
```



substitution

2

*

4

>=

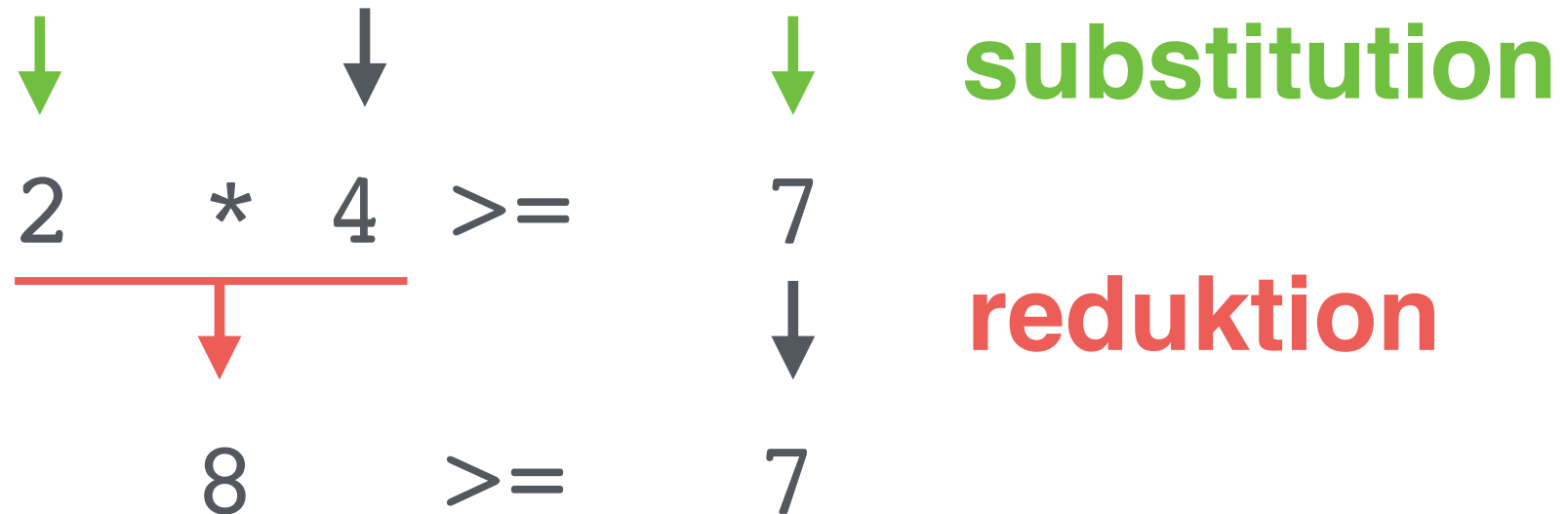
7

Expressions med relationelle operatorer

```
cars = 2
```

```
people = 7
```

```
print(cars * 4 >= people)
```

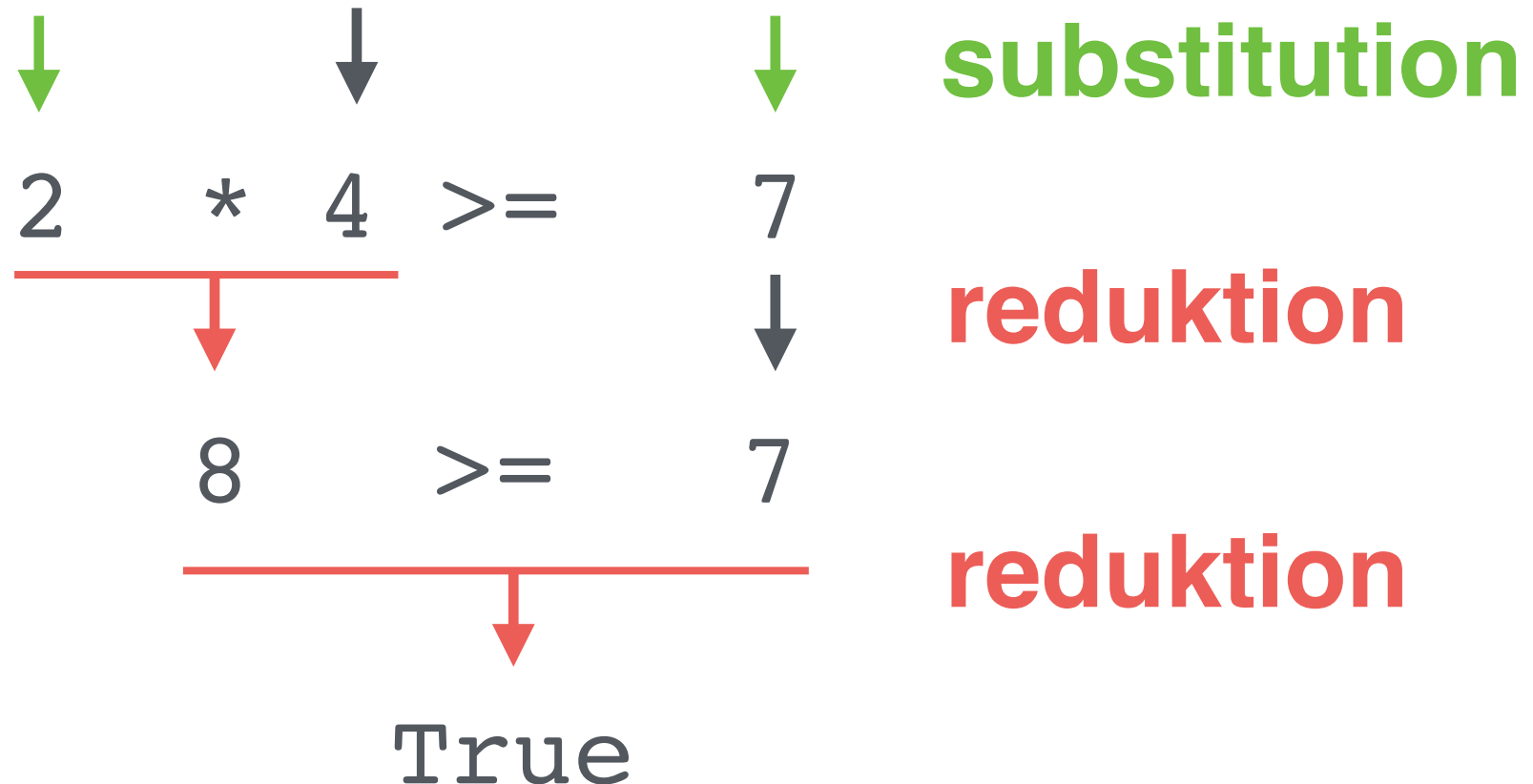


Expressions med relationelle operatorer

```
cars = 2
```

```
people = 7
```

```
print(cars * 4 >= people)
```



Flere expressions

```
print(7 * 6 >= 7)
```

```
print(1 * 2 != 1 + 1)
```

```
print(2 == "2")
```

```
print("A" < "a")
```

```
print("2" < "A")
```

```
print(2 < "2")
```

Logiske operatører

not

Laver sandt til falsk, og falsk til sandt

Operator precedence

Level	Category	Operators
Highest	exponent	**
	multiplication	*, /, //, %
	addition	+, -
	relational	!=, ==, <=, >=, <, >
Lowest	logical	not

not

not 7 > 3

not 2 * 4 == 3 * 5

not True

not False



not

not 0

not 2

not -1

not 3.1415

not 'Banana'

not ''

Logiske operatører

and or

Lader os kombinere expressions, der reducerer til noget sandt eller falsk

Operator precedence

Level	Category	Operators
Highest	exponent	**
	multiplication	*, /, //, %
	addition	+, -
	relational	!=, ==, <=, >=, <, >
	logical	not
Lowest	logical	and
	logical	or



and

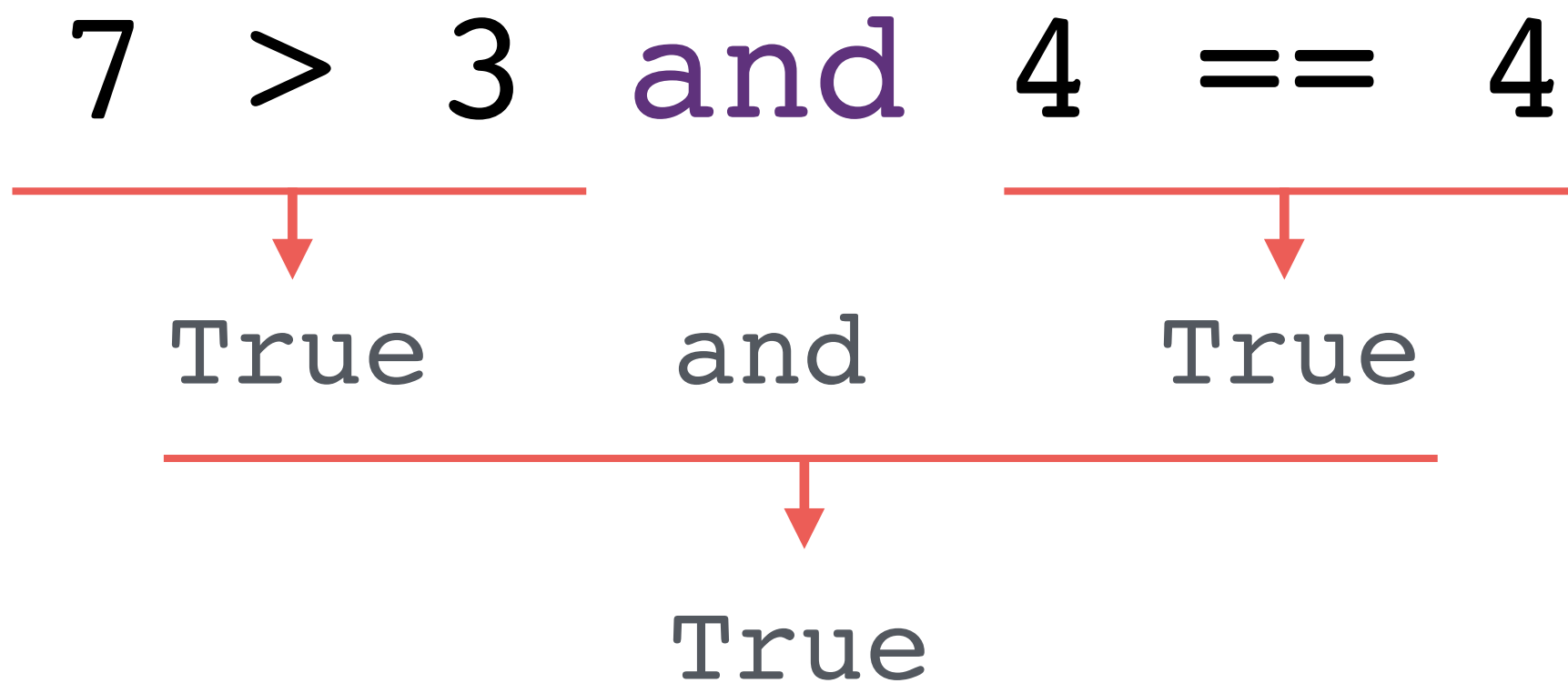
7 > 3 and 4 == 4

and

$7 > 3$ and $4 == 4$
True and True

Reduktion

and



Reduktion

Reduktion



and

True and True

False and True

True and False

False and False



and

```
print(True and 1)
```

```
print(0 and True)
```

```
print(True and 0)
```

```
print(0 and False)
```

and

True and True

False and True

True and False

False and False



or

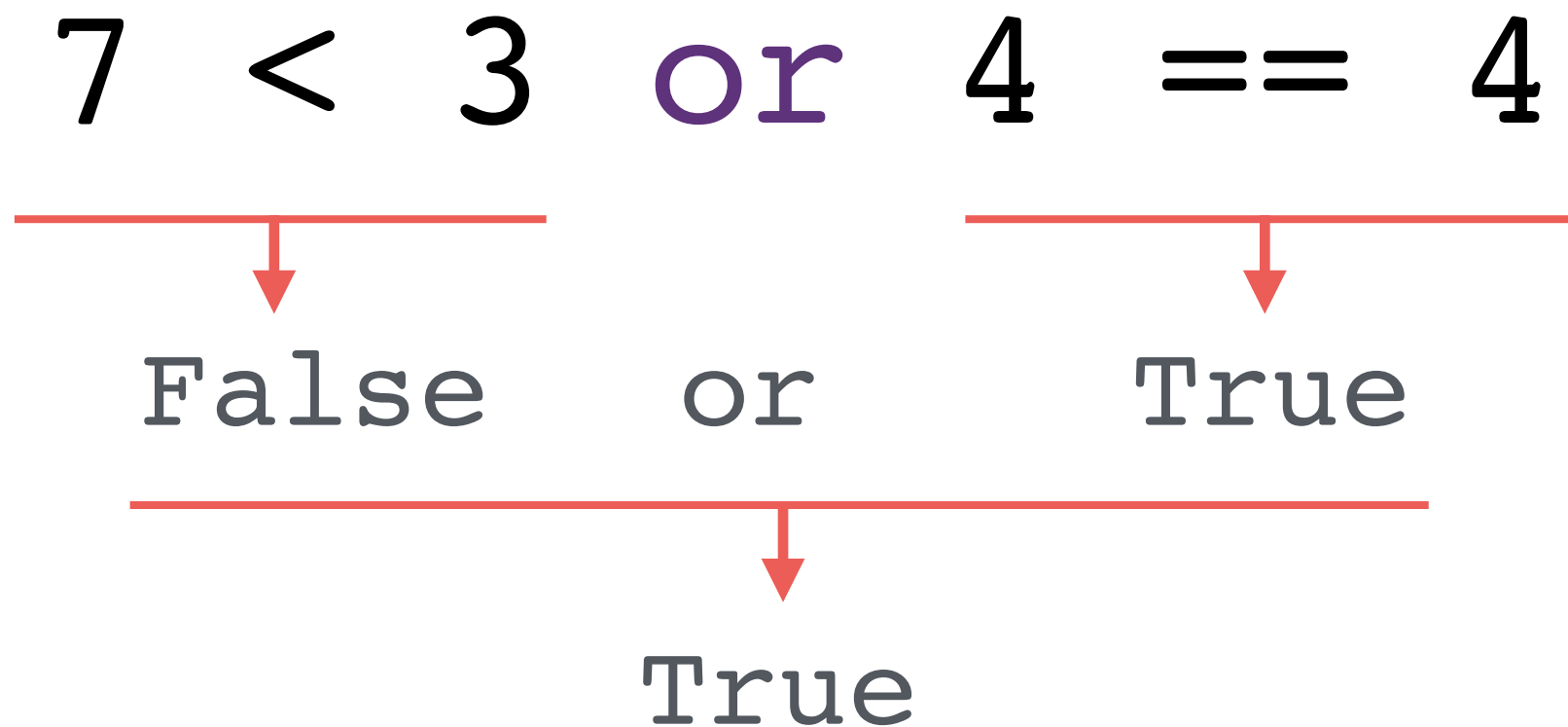
7 < 3 or 4 == 4

or

$7 < 3$ or $4 == 4$
False or True

Reduktion

or



Reduktion

Reduktion



or

True or True

False or True

True or False

False or False

or

True or True

False or True

True or False

False or False



or

```
print(1 or True)
```

```
print(False or 1)
```

```
print(1 or False)
```

```
print(False or 0)
```



Udfordring

```
x = 7
```

```
y = 4
```

```
print(x > y and x or y)
```

What is printed? 7 or 4?